Generative Transcoding of Communicative UI Cues

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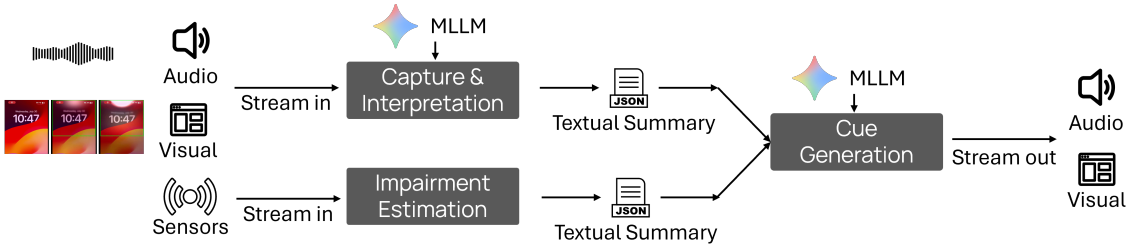

Fig. 1. Existing UIs often rely on static communicative cues (e.g., fixed animations or sounds) that often fail when situational impairments block the intended sensory channel. Generative Transcoding framework addresses this by: (1) detecting the communicative intent of dynamic events from the audio-visual stream, (2) estimating the user's real-time sensory availability, and (3) dynamically generating compensatory cues (e.g., transcoding a visual loading spinner into a audio pulse) to ensure reliable information delivery.

1 Introduction

User interfaces have evolved from a collection of static widgets into dynamic, living designs. Modern applications communicate critical state changes and information not just through text labels, but through a rich, diverse combination of multimodal cues. For example, a clicking sound to indicate the tap-to-pay has gone through, and a flashing red boundary on the phone unlock view to indicate a failed attempt. These signals are often subtle and fleeting, relying on the user's full sensory attention to convey their meaning. However, these pre-determined designs rely on an assumption that the user's context matches the designer's intent. In the real world, users operate in dynamic environments where situational impairments, such as high ambient noise, divided visual attention, or bright sunlight, can reduce or invalidate the information communicated through these channels. Especially in cases where the information is delivered through a single channel only, it is likely that the user may miss these communicative UI cues.

In this paper, we argue that the promise of generative UI lies not just in creating static layouts, but in the runtime generation of dynamic communicative cues. Based on this promise, we propose a framework for Generative Transcoding, a system where multimodal AI models acts as a semantic proxy. Unlike existing accessibility tools that parse static DOM elements, this proxy actively monitors the application's video and audio output stream to capture and interpret the communicative intent of dynamic events. It then estimates the user's sensory availability via mobile sensors and dynamically generates compensatory multimodal cues, such as generating a flashing green boundary to compensate for the clicking confirmation sound, to reduce the impact of user's situational constraints on the effectiveness of such communicative cues. Below we discuss the overall framework to achieve generative transcoding, and report preliminary evidence to show the capability of existing MLLMs on interpreting visual communicative cues and generating communicative cues from extracted intents and situational constraints.

2 Generative Transcoding

As shown in Figure 1, the framework consists of 3 major components, which include

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Intent Capture and Interpretation: This module monitors the raw video and audio output stream of the interface, and uses a MLLM to identify and interpret communicative cues from the input stream. Specifically, it classifies the type of the cues using existing taxonomies, and provides a textual description of its meaning. Our preliminary evaluation shows that existing state-of-the-art MLLMs such as GPT-o3 can achieve an average of 3.47 out of 5 semantic similarity against human interpretations on animated visual cues, and a 62% accuracy on categorizing animation cue purpose (N=300). While performance gaps exists, these results indicate that off-the-shelf MLLMs demonstrate sufficient reasoning capabilities to decode the "communicative intent" of complex UI animations without task-specific training.

Impairment Estimation: To determine the optimal output channel, this module estimates the user's sensory and attention availability from mobile sensor readings, including microphone, IMU, camera, and built-in gaze tracking. Building on prior work in context-aware sensing from mobile devices [6] and LLM-driven inference of human input/output channels [5], the system detects situational impairments, such as "induced deafness" from noise or "induced blindness" from motion in real-time, and creates a dynamic map of available modalities.

Cue Generation: This module compensates for blocked sensory channels by generating alternative and/or additional feedback streams. It utilizes interpreted UI intent and estimated impairment to generate new cues that align with the user's current availability. Specifically, this module should support generation of commonly defined categories from prior taxonomy [1–4], and is adjustable along certain dimensions for flexibility (e.g., level of details, level of intensity).

3 Considerations for Generative Cues

Compared to layout generation, generating dynamic cues has its unique challenges, which we discuss below.

Parametric Generation vs. Open-Ended Synthesis: While technically feasible, using MLLMs or generative models to synthesize raw audio, animations, or UI code introduces unpredictability, which negatively impacts user experience. A UI cue must be recognizable and consistent. Therefore, we argue for a constrained generation framework. Rather than synthesizing assets directly, the LMM should act as a parametric engine that retrieves and modifies high-quality UI primitives. In this view, cue generation is treated as a code modification task: the model identifies a semantic intent (e.g., "Critical Warning") and modulates specific parameters of a pre-defined primitive, such as Salience (haptic strength), Fidelity (TTS vs. Earcon), and modality-dependent variables (visual variables) to accommodate the context. This approach ensures that the output remains explainable, controllable, and semantically stable.

Salience vs. Intrusiveness: The fundamental goal of a cue is to solicit attention, but the cost of attention varies by context. The generative model must dynamically balance perceptual salience (detectability) against cognitive intrusiveness (distraction). A "High Urgency" event (e.g., system failure) requires high-intensity generation. Conversely, a "Low Urgency" status update (e.g., background download) requires ambient generation that requires less perception (e.g., a slow-breathing light or sub-threshold vibration). The system must effectively calculate an "Attention Budget" based on the user's current cognitive load and tune the generated cue's magnitude accordingly.

Optimizing towards existing UI: . Unlike generating a standalone app, transcoding involves overlaying new signals into a host interface. This requires non-destructive integration. The generated cue must achieve visual and functional harmony with the original UI, ensuring it does not occlude critical interaction zones (e.g., covering a "Confirm" button with a generated flash) or conflict with existing UI cues. The generative process must be context-aware not just of the user, but of the application state and the UI layout.

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